

Justin Lee

(925) 282-9275 | justin.lee91375@gmail.com | justinl.me | github.com/jcl23 | [LinkedIn](#) | Lafayette, CA |

CERTIFICATIONS

AWS Certified Cloud Practitioner (2026) | **AWS Certified Solutions Architect** (*In Progress*)

SKILLS

Languages: JavaScript, TypeScript, HTML/CSS, C#, Java, SQL, MATLAB, Python, Haskell, C, C++
Cloud: AWS (Lambda, DynamoDB, API Gateway, Cognito, IAM), Terraform
Databases: MongoDB, EntityFramework, Microsoft SQL Server
Software: Linux, Git, VSCode, Android Studio, IntelliJ IDEA, MATLAB, LaTeX, Jupyter
AI Tools: Claude Code, GitHub Copilot — AI pair programming as a core part of the development workflow
Additional: JS: three.js, Node.js, React, React Native, Angular, MathJax Python: numpy, PyTorch, Matplotlib, OpenCV C#: .NET 3.0, .NET Core, MVC, MVVM

TECHNICAL WORK

Tactic Coordination App | *TypeScript, React, AWS Lambda, DynamoDB, OpenAI API, Claude Code* Nov 2025 - May 2026

- Lets players coordinate tactics in real time — create or join matches via short code, with live WebSocket updates and match state persisted across reconnects.
- Dual auth via Cognito OAuth and Steam OpenID with JWT-based WebSocket session authorization; serverless infrastructure managed with Terraform.
- Integrated OpenAI API and local SLMs to parse natural language player instructions into structured graphs of verifiable match objectives. [Repository](#)

Homology Calculator: Learn Topology Interactively | *TypeScript, React, three.js, GitHub Copilot* Oct 2024 - Mar 2025

- Editor for constructing topological spaces via cell composition and gluing; computes and displays H_0 , H_1 , H_2 homology groups and Betti numbers in a live table.
- Lets users build classical spaces (torus, Klein bottle, sphere) and verify algebraic topology properties hands-on, bridging theory and intuition. [Repository](#) / [Live Demo](#)

Rotation Symmetry Navigator | *TypeScript, React, React-spring, three.js* Dec 2023 - Sep 2024

- 3D visualization of rotation symmetry groups for classical solids (cube, tetrahedron, icosahedron), with animated transitions between symmetry states.
- Includes Cayley diagrams and multiplication tables for all three classical rotation groups (T, O, I), covering the five Platonic solids. [Repository](#) / [Live Demo](#)

MiniJava Compiler | *Java* May 2024

- Full compiler pipeline for a Java subset: lexer, parser, name/type analysis, JVM bytecode emission, and constant folding.
- Supports classes, inheritance, integer arrays, and recursive methods; outputs valid JVM class files executable on the Java runtime.

Workforce Management App | *React Native, Firebase* May 2023

- iOS/Android app for a regional painting business: admin-to-field job dispatch and digital hour logging, replacing paper-based tracking.
- Firebase backend provides real-time data sync and auth; concurrent admin and worker sessions update live without manual refresh.

PROFESSIONAL EXPERIENCE

Software Developer Intern | *Microchip, Inc* | *C#, .NET 3.0, MVC, MVVM* Jun 2023 - Dec 2023

- Full-stack dashboard for real-time plant performance and downtime monitoring, used by engineering and operations staff.
- Redesigned database schema to consolidate plant metrics previously spread across separate manual systems; maintained read availability during live data updates.
- On-site QC tool for technicians to log and submit chip defect scores and impact ratings, replacing manual data entry.

Software Developer Intern | *Supra, a Carrier company* | *C#, .NET 3.0, .NET Core* Jun 2022 - Dec 2022

- Designed REST API endpoints for Bluetooth lock access control and authorization, shipped to customers on commercial Supra lockbox hardware.
- Python microservice for cross-codebase localization updates; reduced turnaround by 98%.
- .NET desktop tool for QA engineers to track device failure/success rates across physical security hardware.

EDUCATION

Portland, OR **BA in Computer Science, Math minor** | University of Portland *Graduation date: May 2024*